

UC Test Case Dataset Summary

Seven Network Unit Commitment Instances for Algorithm Scaling Studies

Topology: IEEE benchmarks | UC parameters: Kazarlis (1996), RTS-GMLC (2019), UCjl (2020)

1. Dataset Overview

Seven network UC instances are provided, covering a 50× range in system size (6 to 300 buses) and a 80× range in peak demand (283 MW to 23,525 MW). The first three cases (6-bus, IEEE 14, IEEE 39) form the primary experimental set. The remaining four (IEEE 30, 57, 118, 300) form the scaling study set.

Test case	Buses	Lines	Gens	RES	Peak demand	Installed cap.	RES cap.	Topology source	UC params source
6-bus (Wood & Wollenberg)	6	9	3	2	286 MW	430 MW	280 MW	Wood & Wollenberg (1996)	Kazarlis (1996)
IEEE 14-bus	14	20	6	2	648 MW	902 MW	570 MW	Christie (1993)	Kazarlis (1996) + RTS-GMLC
IEEE 39-bus (New England)	39	46	10	4	2,656 MW	7,010 MW	1,900 MW	Athay (1979), Pai (1989)	Kazarlis (1996) + RTS-GMLC
IEEE 30-bus	30	41	6	2	283 MW	435 MW	74 MW	Christie (1993)	UCjl / Xavier (2020)
IEEE 57-bus	57	76	7	3	1,251 MW	1,180 MW	276 MW	Christie (1993)	UCjl / Xavier (2020)
IEEE 118-bus	118	120	54	4	4,242 MW	12,885 MW	986 MW	Christie (1993)	UCjl / Xavier (2020)
IEEE 300-bus	300	309	69	5	23,525 MW	16,195 MW	2,908 MW	IEEE Task Force (1993)	UCjl / Xavier (2020)

RES = Renewable Energy Sources (wind + solar PV). Installed cap. = sum of all generator Pmax. All cases use T=24 h horizon, carbon price = 50 \$/tCO₂, VoLL = 20,000 \$/MWh.

2. Parameter Methodology

2.1 Network topology and generator capacities

The network topology (buses, lines, susceptances, thermal limits) and generator capacity limits ($P_{\min}$, $P_{\max}$) are taken verbatim from the standard IEEE benchmark sources listed in Section 5. No modifications are made to these elements.

2.2 UC parameters — primary cases (6-bus, IEEE 14, IEEE 39)

Thermal generator UC parameters are drawn from Kazarlis et al. (1996) [1], the canonical 10-unit UC benchmark, matched by technology type and capacity class. Costs are scaled linearly by $P_{\max} / P_{\max_ref}$. Nuclear unit parameters follow RTS-GMLC / Barrows et al. (2019) [2]: $UT = DT = 168$ h, startup cost = $500 \text{ \$/MW} \times P_{\max}$. Hydro parameters also follow RTS-GMLC [2]: $UT = DT = 1$ h, zero startup/shutdown/no-load costs.

2.3 UC parameters — scaling cases (IEEE 30, 57, 118, 300)

UC parameters for the scaling cases follow the methodology of UnitCommitment.jl (Xavier et al., 2020) [3], which is published under a modified BSD license with a citable Zenodo DOI. The specific rules applied are:

- Ramp-up, ramp-down, startup rate, and shutdown rate: set to 30% of $P_{\max}$ per hour (UCjl fixed proportion rule).
- Minimum up/down times (UT/DT): assigned by technology class using Kazarlis type mapping (coal large: 8/8 h, coal mid: 6/6 h, CCGT: 3/3 h, OCGT/diesel: 1/1 h).
- Fuel costs: assigned per technology type consistent with Kazarlis cost ordering.
- No-load, startup, and shutdown costs: scaled from Kazarlis reference values by $P_{\max}$ ratio.

2.4 Demand profiles

All demand profiles are synthetic 24-hour normalized load curves following a standard diurnal shape, scaled to the system peak of each benchmark. Bus-level demand is distributed proportionally to the base-case MW loads published in MATPOWER.

2.5 Renewable units

Renewable units (wind and solar PV) are added to each case at selected buses. Wind availability follows a diurnal profile consistent with NREL ATB archetypes [4]. Solar PV follows a standard bell-curve irradiance profile. Curtailment cost is set to 5 $\text{\$/MWh}$ for all units.

3. Case-by-Case Summaries

3.1 6-bus (Wood & Wollenberg)

The smallest case, used for rapid prototyping and verification. Three thermal generators spanning a low-cost coal, a mid-cost CCGT, and a peaking OCGT. Two renewable units (wind at bus 2, solar at bus 5) provide meaningful curtailment dynamics despite the small network size. All UC parameters cited from Kazarlis [1] directly.

Parameter	Value
Buses / Lines	6 / 9
Thermal generators	3 (1 Coal, 1 CCGT, 1 OCGT)
Renewable units	2 (Wind 180 MW, Solar 100 MW)

Peak system demand	286.4 MW
Total installed capacity	430 MW
Renewable penetration (peak)	98.5%
UC param source	Kazarlis et al. (1996) [1] — direct
Topology source	Wood & Wollenberg (1996) [5]

3.2 IEEE 14-bus

A small but diverse system with six thermal units spanning four technology types. The inclusion of a nuclear unit (bus 7, 200 MW) and a peaking diesel (bus 12, 50 MW) creates interesting commitment dynamics. Two large renewables (350 MW wind, 220 MW solar) exceed total thermal peak at some hours.

Parameter	Value
Buses / Lines	14 / 20
Thermal generators	6 (2 Coal, 1 CCGT, 1 OCGT, 1 Nuclear, 1 Diesel)
Renewable units	2 (Wind 350 MW, Solar 220 MW)
Peak system demand	648.4 MW
Total installed capacity	902 MW
Renewable penetration (peak)	87.9%
UC param source	Kazarlis [1] for thermal; RTS-GMLC [2] for nuclear
Topology source	Christie (1993), PSTCA [6]

3.3 IEEE 39-bus (New England)

The largest of the primary cases, with a rich generation mix including two nuclear baseload units, two hydro units, three CCGT units, and three coal units. Four renewable units (total 1,900 MW peak) create significant congestion and curtailment scenarios across the 46-line network.

Parameter	Value
Buses / Lines	39 / 46
Thermal generators	10 (3 Coal, 3 CCGT, 2 Nuclear, 2 Hydro)
Renewable units	4 (3 Wind, 1 Solar — total 1,900 MW peak)
Peak system demand	2,656.4 MW
Total installed capacity	7,010 MW
Renewable penetration (peak)	71.5%
UC param source	Kazarlis [1] for thermal; RTS-GMLC [2] for nuclear & hydro
Topology source	Athay et al. (1979) [7], Pai (1989) [8]

3.4 IEEE 30-bus (scaling)

Entry point for the scaling study. Six generators of mixed type on a 41-line network. System peak is comparable to the 6-bus case but with five times the network complexity. Two renewable units (wind + solar, 74 MW combined peak) provide moderate renewable penetration.

Parameter	Value
Buses / Lines	30 / 41
Thermal generators	6 (2 Coal-mid, 2 CCGT, 2 OCGT)
Renewable units	2 (Wind 60 MW, Solar 40 MW)
Peak system demand	283.4 MW
Total installed capacity	435 MW
Renewable penetration (peak)	36.1%
UC param source	UCjl methodology — Xavier et al. (2020) [3]
Topology source	Christie (1993), PSTCA [6]

3.5 IEEE 57-bus (scaling)

Mid-size scaling case. Seven generators on a 76-line network with a 1,251 MW peak demand. Three renewable units (two wind, one solar, 276 MW combined) provide roughly 22% renewable penetration at peak. The MATPOWER case57 fmax values are used from the pglib-opf corrected version.

Parameter	Value
Buses / Lines	57 / 76
Thermal generators	7 (3 Coal, 2 CCGT, 2 OCGT)
Renewable units	3 (2 Wind, 1 Solar — total 276 MW peak)
Peak system demand	1,251.0 MW
Total installed capacity	1,180 MW
Renewable penetration (peak)	22.1%
UC param source	UCjl methodology — Xavier et al. (2020) [3]
Topology source	Christie (1993), PSTCA [6]

3.6 IEEE 118-bus (scaling)

Large-scale case with 54 generators and 120 lines. The 54-generator fleet creates a rich UC problem. Four renewable units (two wind, two solar, 986 MW combined) provide approximately 23% penetration. This case is the standard in recent scalability papers for network UC.

Parameter	Value
Buses / Lines	118 / 120
Thermal generators	54 (classified by Pmax into CCGT/OCGT/Diesel)
Renewable units	4 (2 Wind, 2 Solar — total 986 MW peak)

Peak system demand	4,242.0 MW
Total installed capacity	12,885 MW
Renewable penetration (peak)	23.2%
UC param source	UCjl methodology — Xavier et al. (2020) [3]
Topology source	Christie (1993), PSTCA [6]

3.7 IEEE 300-bus (scaling)

The largest case in the dataset. 69 generators on a 309-line network with a 23,525 MW system peak — approximately 90× larger than the 6-bus case. Five renewable units (three wind, two solar, 2,908 MW combined) provide 12% renewable penetration. This case tests algorithm scalability at transmission-system scale.

Parameter	Value
Buses / Lines	300 / 309
Thermal generators	69 (mix of coal-large, CCGT, OCGT)
Renewable units	5 (3 Wind, 2 Solar — total 2,908 MW peak)
Peak system demand	23,525.0 MW
Total installed capacity	16,195 MW
Renewable penetration (peak)	12.4%
UC param source	UCjl methodology — Xavier et al. (2020) [3]
Topology source	IEEE Task Force (1993) [9]

4. Ready-to-Paste Paper Section

We evaluate the proposed method on seven network UC instances covering a 50x range in system size. Three primary cases — the 6-bus system [5], IEEE 14-bus [6], and IEEE 39-bus New England system [7,8] — are used for detailed analysis. Four additional cases — IEEE 30-bus, 57-bus, 118-bus, and 300-bus [6] — form a scaling study set. Network topology, line susceptances, thermal limits, and generator capacity limits are taken verbatim from the respective benchmark sources. UC operational parameters for the primary cases are drawn from the 10-unit benchmark of Kazarlis et al. [1], matched by technology type and scaled proportionally to Pmax; nuclear and hydro parameters follow the RTS-GMLC test system [2]. UC parameters for the scaling cases follow the methodology of UnitCommitment.jl [3], setting ramp rates to 30% of Pmax per hour and assigning minimum up/down times by technology class. Renewable units are added at selected buses with diurnal availability profiles following NREL ATB archetypes [4]. All demand profiles are 24-hour normalized load curves scaled to each system's peak.

5. References

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